

MUKESH SABESAN

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Education

Toronto Metropolitan University

Bachelor of Engineering in Computer Engineering

Graduation Date: April 2026

Dean's List: 2023-24, 2025-26

Experience

Computer Engineering Intern

May 2024 - August 2025

Ontario Power Generation

Pickering, ON

- Earned the **OPG Power of You Award** Nomination for work on the Pickering Nuclear Generating Station (PNGS) Defuel Software Modification project, resulting in **time savings of over 75 days** and a reduction in cancellations reported by Authorized Nuclear Operators.
- Authored and revised comprehensive Commercial Grade Dedication hardware test plans for **4** computer and control systems at PNGS, supporting the reliable operation of the PNGS Digital Control Computers and power generation systems, powering **1.5 million** homes daily and generating **\$4-6 million** in revenue daily for OPG.
- Developed a **Python-based automation script** to monitor the progress of the PNGS A Safe Storage project, enabling efficient tracking of defueling progress and errors for a **\$500+ million project**.
- Compiled editable excel sheets of **5000+** rows of Input/Output lists (IOs) and annunciations, optimizing the monitoring of IOs and annunciations during the stabilization and storage phases of the PNGS A Safe Storage project.
- Assisted Project Managers and Senior Engineers by managing documentation and material tracking throughout the project lifecycle, ensuring stakeholder alignment and timely completion of deliverables.
- Earned **9** formal recognitions from **6** managers for key contributions to the PNGS A Safe Storage project, Defuel Software modification, Auxiliary Power System and ensuring the reliable operation of Pickering Nuclear Generating Station.

Project Manager

September 2023 - Dec. 2023, September 2025 - Dec. 2025

Faculty of Engineering and Architectural Science at TMU

Toronto, ON

- Facilitated and managed teams of **30+** team engineering students in the successful planning and execution of multiple projects, ensuring effective collaboration.
- Mentored peers and modeled strong engineering practices to promote academic excellence.
- Coordinated every aspect of project execution, from conception to completion, enabling a student team to advance in the TMU Student Design Competition through focused task management and continuous progress evaluation.

Recreation Facilities and Fitness Centre Assistant

September 2023 - April 2024

Toronto Metropolitan University Recreation

Toronto, ON

- Provided excellent customer service to TMU's **world-class** RAC and MAC athletic facilities members at TMU by welcoming members, advertising tours of the facilities and addressing customers various needs and concerns.
- Conducted evacuations for **3 Fire Alarm drills/Actual Fire Alarms** as the Fire Warden at the RAC, adhering to the fire safety protocols and ensuring Security and First Responders can respond appropriately and promptly.

Projects

Network Based System-On-Chip (SoC) (Capstone) | *System-Verilog, AMD-Xilinx* September 2025 - April 2026

- Designed and Developed a Network-on-Chip router core for an NoC-based System-on-Chip Programmable Chip (SoPC) for Image Processing.
- Integrated the Sobel Edge Detection and Inversion image processing filters for the 4 filters on the NoC.

Fitness Centre Fullstack Application (FitHub) | *React, Next, TypeScript* | [GitHub Repo](#) December 2025

- Designed a full-stack application for a Fitness Centre to manage members, classes and gym operations.
- Developed the Member Dashboard and Profile Management system of FitHub, where members can view their upcoming classes, update and add their fitness goals, and update and view their profile and membership information.

Smart Parking Management System (SpotFree) | *React, Python, JavaScript, Docker* | [GitHub Repo](#) April 2026

- Developed a Smart Parking System with a React front-end, FastAPI backend services, PostgreSQL, and RabbitMQ, and containerized with Docker.
- Integrated the reservation service of the application, allowing user's reservations to update in real-time across services in the application.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, VHDL, C, C#, C++, SQL, Assembly Programming, HTML, CSS, MATLAB, System-Verilog

Tools/Technologies: Git/Github, PostgreSQL, Docker, Kubernetes, NI Multisim, Quartus II, UNIX, Power BI, Power Apps, SharePoint, Microsoft Office Suite, KiCAD, AutoCAD, VMWare, AMD Xilinx ISE, Firebase